Lab 4.1 - Configure VLANs and Trunking

Important for Online Workshop Students Only:

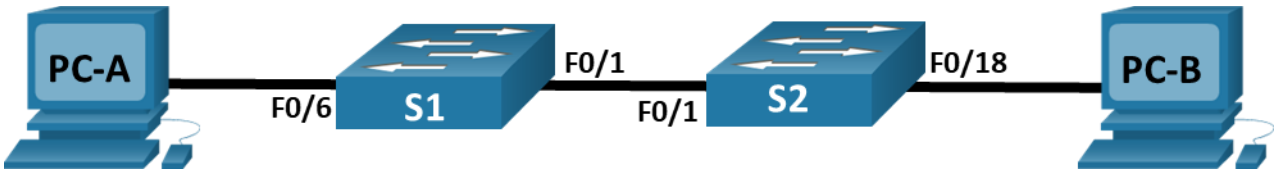
You must use Packet Tracer v8.2.2 to complete this lab.

Please rename your Packet Tracer file to FirstName_LastName_sNumber_Lab 4.1.pka

Marking Scheme

No.		Marks
1	PC-A	6
2	PC-B	6
3	VLAN settings for S1 configured correctly – Part 2 and Part 4	34
4	VLAN settings for S2 configured correctly – Part 2 and Part 4	34

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	192.168.1.11	255.255.255.0	N/A
S2	VLAN 1	192.168.1.12	255.255.255.0	N/A
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.10.4	255.255.255.0	192.168.10.1

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Create VLANs and Assign Switch Ports

Part 3: Maintain VLAN Port Assignments and the VLAN Database

Part 4: Configure an 802.1Q Trunk between the Switches

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Background / Scenario

Modern switches use virtual local-area networks (VLANs) to improve network performance by separating large Layer 2 broadcast domains into smaller ones. VLANs can also be used as a security measure by controlling which hosts can communicate. In general, VLANs make it easier to design a network to support the goals of an organization. Communication between different VLANs requires a device operating at Layer 3 of the OSI model.

VLAN trunks are used to span VLANs across multiple devices. Trunks allow the traffic from multiple VLANs to travel over a single link, while keeping the VLAN identification and segmentation intact.

In this lab, you will create VLANs on both switches in the topology, assign VLANs to switch access ports, verify that VLANs are working as expected, and then create a VLAN trunk between the two switches to allow hosts in the same VLAN to communicate through the trunk, regardless of which switch the host is actually attached to.

Note: For students in on-campus workshops only: Make sure that the switch has been erased and has no startup configurations. Please refer to Appendix A

Note: For students in online workshops only: In Packet Tracer, you must use the 2960 switch

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings on the PC hosts and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each switch.

a. Assign a device name to the switch.

b. Disable DNS lookup to prevent the switch from attempting to translate incorrectly entered commands as though they were host names.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the vty password and enable login.

f. Configure logging synchronous for the console line and vty lines.

g. Encrypt the plaintext passwords using the service password-encryption command:

```
S1(config)# service password-encryption
```

h. Create a banner that warns anyone accessing the device that unauthorized access is prohibited. The banner text must be exact:

```
S1(config)# banner motd 'Unauthorised access is prohibited and will be
            prosecuted strictly.'
```

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i. Configure the switch for SSH access including the vty lines. Use username **admin** and password **cisco**.

```
S1(config)# ip domain-name networklab.com
S1(config)# crypto key generate rsa
      The name for the keys will be: S1.networklab.com
      Choose the size of the key modulus in the range of 360 to 2048
      for your General Purpose Keys. Choosing a key modulus greater
      than 512 may take a few minutes.

      How many bits in the modulus [512]: 1024

      % Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
      (elapsed time was 1 seconds)
      *Jan 28 21:09:29.867: %SSH-5-ENABLED: SSH 1.99 has been enabled
S1(config)# username admin privilege 15 secret cisco
S1(config)# line vty 0 15
S1(config-line)# transport input ssh
S1(config-line)# login local
S1(config-line)# exit
```

Note: Because the switches are on a different network to the PCs, you will not be able to use the terminal to connect to the switch using SSH despite configuring SSH. You will need to use the console connection for all configuration for this activity. This behaviour is intended for this lab. To allow the connection would require some form of routing between the networks. That will come later in the course.

j. Configure the IP address listed in the Addressing Table for VLAN 1 on the switch.

```
S1#(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.11 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
```

Note: For the Packet Tracer version of this lab, the score will not change after setting VLAN 1 IP address here as we will change it later in Step 2 and the final configuration will attract the scoring points.

k. Administratively deactivate all unused ports on the switch by using the **shutdown** command on those ports.

Eg. For Switch shutdown ports 2-5, and 7-24 as ports 1 and 6 are in use:

```
S1(config)# interface range f0/2-5
S1(config-if-range)# shutdown
S1(config-if-range)# end
S1(config)# interface range f0/7-24
S1(config-if-range)# shutdown
S1(config-if-range)# end
```

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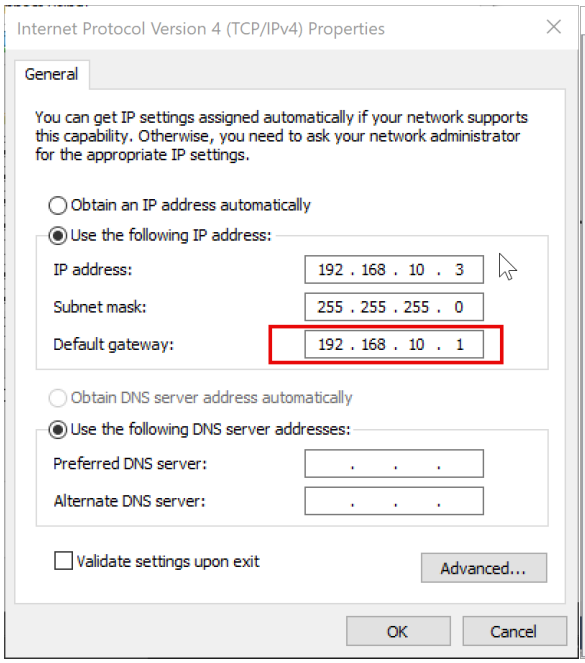
For Switch S2, using similar commands to the above, but turn off the corresponding unused ports for Switch S2.

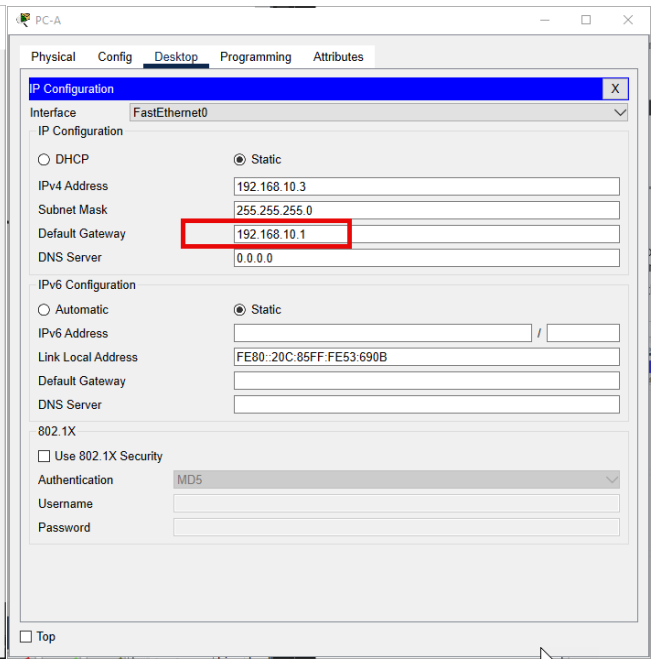
Step 3: Configure PC hosts.

Refer to the Addressing Table for PC host address information to configure the IP address, subnet mask, and default gateway. In the configuration from the addressing table, the PCs are on the same network.

Note: the default gateway is entered in the same place as the IP address and subnet mask (shown here are the physical device and the packet tracer IP configuration windows):



Physical device



Packet Tracer device

Step 4: Test connectivity.

Verify that the PC hosts can ping one another.

Note: It may be necessary to disable the PC's firewall to ping between PCs (for students in on-campus workshops only).

Can PC-A ping PC-B? _____

Can PC-A ping S1? _____

Can PC-B ping S2? _____

Can S1 ping S2? _____

If you answered no to any of the above questions, why were the pings unsuccessful?

_____

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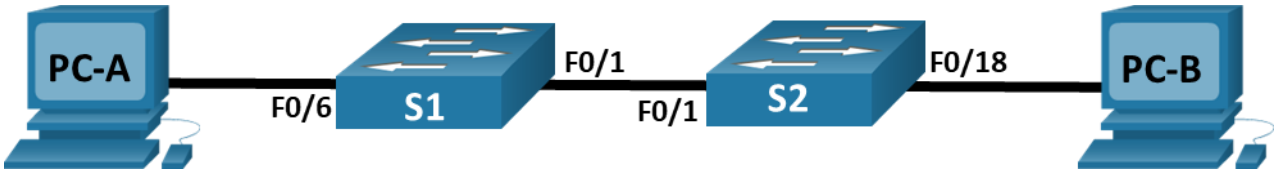
Part 2: Create VLANs and Assign Switch Ports

Topology



Updated Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 99	192.168.1.11	255.255.255.0	N/A
S2	VLAN 99	192.168.1.12	255.255.255.0	N/A
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.10.4	255.255.255.0	192.168.10.1

In Part 2, you will create **VLANs** called **Management**, **Operations**, and **Parking_Lot**, as well as the **Native VLAN** on both switches. You will then assign the VLANs to the appropriate interface, this includes moving the IP address for the switch from VLAN 1 (configured in Part 1) to VLAN 99. The updated addressing table above shows the change.

The **show vlan** command is used to verify your configuration settings. Pay particular attention to the change from VLAN 1 to VLAN 99.

Step 1: Create VLANs on the switches.

a. Create the VLANs on S1.

```
S1(config)# vlan 10
S1(config-vlan)# name Operations
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name Parking_Lot
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name Management
S1(config-vlan)# exit

S1(config)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# exit
```

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b. Create the same VLANs on S2.

c. Issue the **show vlan brief** command to view the list of VLANs on S1 and S2.

```
S1# show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gi0/1, Gi0/2
10   Operations                       active
20   Parking_Lot                      active
99   Management                       active
1000 Native                           active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

What is the default VLAN?

_____

What ports are assigned to the default VLAN?

_____

Step 2: Assign VLANs to the correct switch port interfaces.

a. Assign VLANs to the port interfaces on S1 using the updated addressing table in Part 2:

1) Assign PC-A to the Operations VLAN (10):

```
S1(config)# interface f0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# exit
```

2) Move the switch IP address to VLAN 99. Remove the IP address from VLAN 1 and add the IP address to VLAN 99:

```
S1(config)# interface vlan 1
S1(config-if)# no ip address
S1(config-if)# exit

S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.1.11 255.255.255.0
S1(config-if)# end
```

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b. Issue the **show vlan brief** command and verify that the VLANs are assigned to the correct interfaces.

c. Issue the **show ip interface brief** command.

What is the status of VLAN 99? Explain.

_____

d. Assign PC-B to the Operations VLAN on S2 (see point a, item 1 above, for how it was done for PC-A on S1).

e. Remove the IP address for VLAN 1 on S2 (see point a, item 2 above, for how it was done for VLAN 1 on S1).

f. Configure an IP address for VLAN 99 on S2 according to the Addressing Table (see point a, item 2 above, for how it was done for VLAN 99 on S1).

g. Use the **show vlan brief** command to verify that the VLANs are assigned to the correct interfaces.

Is S1 able to ping S2? Explain.

_____

_____

Is PC-A able to ping PC-B? Explain.

_____

_____

Part 3: Maintain VLAN Port Assignments and the VLAN Database

In Part 3, you will change VLAN assignments to ports and remove VLANs from the VLAN database.

Step 1: Assign a VLAN to multiple interfaces.

a. On S1, assign interfaces F0/11 – 24 to VLAN99.

```
S1(config)# interface range f0/11-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 99
S1(config-if-range)# end
```

b. Issue the **show vlan brief** command to verify VLAN assignments.

c. Reassign F0/11 and F0/21 to VLAN 10.

d. Verify that VLAN assignments are correct.

Step 2: Remove a VLAN assignment from an interface.

a. Use the **no switchport access vlan** command to remove the VLAN 99 assignment to F0/24.

```
S1(config)# interface f0/24
S1(config-if)# no switchport access vlan
S1(config-if)# end
```

b. Verify that the VLAN change was made.

Which VLAN is F0/24 now associated with?

_____

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Step 3: Remove a VLAN ID from the VLAN database.

a. Add VLAN 30 to interface F0/24 without issuing the global VLAN command.

```
S1(config)# interface f0/24
S1(config-if)# switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
```

Note: Current switch technology no longer requires that the **vlan** command be issued to add a VLAN to the database. By assigning an unknown VLAN to a port, the VLAN will be created and added to the VLAN database.

b. Verify that the new VLAN is displayed in the VLAN table.

What is the default name of VLAN 30?

_____

c. Use the **no vlan 30** command to remove VLAN 30 from the VLAN database.

```
S1(config)# no vlan 30
S1(config)# end
```

d. Issue the **show vlan brief** command. F0/24 was assigned to VLAN 30.

After deleting VLAN 30 from the VLAN database, what VLAN is port F0/24 assigned to? What happens to the traffic destined to the host attached to F0/24?

_____

_____

_____

e. Issue the **no switchport access vlan** command on interface F0/24.

f. Issue the **show vlan brief** command to determine the VLAN assignment for F0/24.

To which VLAN is F0/24 assigned?

_____

Note: Before removing a VLAN from the database, it is recommended that you reassign all the ports assigned to that VLAN.

Why should you reassign a port to another VLAN before removing the VLAN from the VLAN database?

_____

_____

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Part 4: Configure an 802.1Q Trunk Between the Switches

In Part 4, you will configure interface F0/1 to use the Dynamic Trunking Protocol (DTP) to allow it to negotiate the trunk mode. After this has been accomplished and verified, you will disable DTP on interface F0/1 and manually configure it as a trunk.

Step 1: Use DTP to initiate trunking on F0/1.

The default DTP mode of a 2960 switch port is dynamic auto. This allows the interface to convert the link to a trunk if the neighboring interface is set to trunk or dynamic desirable mode.

a. Set F0/1 on S1 to negotiate trunk mode.

```
S1(config)# interface f0/1
S1(config-if)# switchport mode dynamic desirable
Sep 19 02:51:47.257: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
Sep 19 02:51:47.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed
state to up
```

You should also receive link status messages on S2.

```
S2#
Sep 19 02:42:19.424: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
Sep 19 02:42:21.454: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed
state to up
Sep 19 02:42:22.419: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
```

b. Issue the **show vlan brief** command on S1 and S2. Interface F0/1 is no longer assigned to VLAN 1. Trunked interfaces are not listed in the VLAN table.

c. Issue the **show interfaces trunk** command to view trunked interfaces. Notice that the mode on S1 is set to desirable, and the mode on S2 is set to auto.

Note: By default, all VLANs are allowed on a trunk. The **switchport trunk** command allows you to control what VLANs have access to the trunk. For now, keep the default settings which allows all VLANs to traverse F0/1.

d. Verify that VLAN traffic is traveling over trunk interface F0/1.

Can S1 ping S2? _____

Can PC-A ping PC-B? _____

Can PC-A ping S1? _____

Can PC-B ping S2? _____

If you answered no to any of the above questions, explain why were the pings unsuccessful.

_____

_____

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Step 2: Manually configure trunk interface F0/1.

The **switchport mode trunk** command is used to manually configure a port as a trunk. This command should be issued on both ends of the link.

a. Change the switchport mode on interface F0/1 to force trunking. Make sure to do this on F0/1 on both switches.

```
S1(config)# interface f0/1
S1(config-if)# switchport mode trunk
```

b. Issue the **show interfaces trunk** command to view the trunk mode. Notice that the mode changed from **desirable** to **on**.

c. Modify the trunk configuration on F0/1 on both switches by changing the native VLAN from VLAN 1 to VLAN 1000.

```
S1(config)# interface f0/1
S1(config-if)# switchport trunk native vlan 1000
```

Note: When you change the native VLAN from 1 to 1000 on Switch 1, you may see the following message letting you know that there is now a mismatch between native VLANs on F0/1 on S1 and F0/1 on S2, as S2 F0/1 still has native VLAN 1 assigned. It is critical to update both the S1 F0/1 and S2 F0/1 native VLANs to fix this mismatch.

```
S1(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer
vlan id 1 on FastEthernet0/1 VLAN1000.


%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN1000.
Inconsistent local vlan.
```

d. Modify the trunk configuration on both switches to specify that only VLANs 10, 20, 99, and 1000 are allowed to cross the trunk.

```
S1(config)# interface f0/1
S1(config-if)# switchport trunk allowed vlan 10,20,99,1000
```

e. Issue the show interfaces trunk command to view the trunk. Notice the Native VLAN information is updated, and the VLANs allowed on the trunk are also updated.

Why would you want to manually configure an interface to trunk mode instead of using DTP?

_____

_____

Why would you want to change the native VLAN on a trunk?

_____

_____

Why would you want to only allow specific VLANs traffic to cross a trunk? What is the problem with leaving it at its default setting, i.e. allow all VLANs traffic?

_____

_____

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Would you still be able to ping PC-B from PC-A if you updated the interface f0/18 configuration on S2 to remove VLAN 10 access and allow VLAN 20 access?

_____

_____

Reflection Questions

1. What is needed to allow hosts on VLAN 10 to communicate to hosts on VLAN 99?

_____

_____

2. What are some primary benefits that an organization can receive through effective use of VLANs?

_____

_____

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Appendix A: Initializing and Reloading a Switch (For Physical Devices Only)

a. Console into the switch and enter privileged EXEC mode.

```
Switch> enable
Switch#
```

b. Use the **show flash** command to determine if any VLANs have been created on the switch.

```
Switch# show flash
Directory of flash:/

    2  -rwx        1919   Mar 1 1993 00:06:33 +00:00  private-config.text
    3  -rwx        1632   Mar 1 1993 00:06:33 +00:00  config.text
    4  -rwx       13336   Mar 1 1993 00:06:33 +00:00  multiple-fs
    5  -rwx    11607161   Mar 1 1993 02:37:06 +00:00  c2960-lanbasek9-mz.150-2.SE.bin
    6  -rwx         616   Mar 1 1993 00:07:13 +00:00  vlan.dat

32514048 bytes total (20886528 bytes free)
```

c. If the **vlan.dat** file was found in flash, then delete this file.

```
Switch# delete vlan.dat
Delete filename [vlan.dat]?
```

d. You are prompted to verify the filename. At this point, you can change the filename or just press Enter if you have entered the name correctly.

e. You are prompted to confirm deleting this file. Press Enter to confirm deletion. (Pressing any other key aborts the deletion.)

```
Delete flash:/vlan.dat? [confirm]
Switch#
```

f. Use the **erase startup-config** command to erase the startup configuration file from NVRAM. You are prompted to confirm removing the configuration file. Press Enter to confirm to erase this file. (Pressing any other key aborts the operation.)

```
Switch# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Switch#
```

g. Reload the switch to remove any old configuration information from memory. You are prompted to confirm reloading the switch. Press Enter to proceed with the reload. (Pressing any other key aborts the reload.)

```
Switch# reload
Proceed with reload? [confirm]
```

Note: You may be prompted to save the running configuration prior to reloading the switch. Type **no** and press Enter.

```
System configuration has been modified. Save? [yes/no]: no
```

h. After the switch reloads, you should be prompted to enter the initial configuration dialog. Type **no** and press Enter.

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>
```

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